Pitch–Hue Correspondences:
A Quantitative Corpus Analysis Spanning Three Millennia of Mapping Systems

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Abstract

Which color is the note C? Newton said red; Rimington said violet. Three centuries of proposals have produced no consensus, and the pattern of that disagreement is itself informative. We present a quantitative corpus analysis of 70 pitch–hue mapping systems, from Chinese *Wuxing* theory to contemporary digital applications, testing four theoretical accounts: structural (shared physics), semantic (shared language), statistical (environmental co-occurrence), and hedonic (shared affective valence). After standardising all specifications to CIELAB and applying CIEDE2000, global agreement across 25 retained systems is low (mean 0.22), close to the simulated chance level of 0.21: the historical record contains no universal pitch–hue mapping. Hue assignments are nonetheless non-uniformly distributed (Rayleigh $R = .66$, $p = .003$), and the disagreement is structured. Organised into eight reasoning families classified from source texts, the corpus shows a consistent within-family ordering: Pedagogical systems agree most (0.348, 57% above the pooled baseline), followed by Synesthetic (0.276), whereas Cosmological systems fall below the pooled baseline-a descriptive pattern that small family sizes leave statistically unresolved. The hedonic account performs best: Procrustes alignment of cross-modal dimensions in Evaluation-Potency-Activity affective space yields a disparity of 0.559 (permutation $p = .027$), with Evaluation as the primary aligning dimension. The structural account earns partial credit (semitone–hue correlation $r = .393$, $p = .002$), but its complementary-hue prediction fails. Sentence-transformer probing recovers pitch–hue geometry under timbral descriptors ($\rho = .368$, $p = .007$), consistent with a semantics–affect entangled channel. Pitch–hue correspondences appear organised by a shared affective geometry, scaffolded by structural and statistical regularities, with implications for a congruency framework for neural research.

Keywords: cross-modal correspondence, pitch–color association, synesthesia, perceptual congruency, affective mediation

Public Significance Statement: Proposals linking musical notes to colors have accumulated for three centuries without agreement. Treating this historical record as data, we show that the disagreement is structured: proposals agree within reasoning traditions, and the traditions align best through the emotional character shared by sounds and colors. The findings give educators, accessibility designers, and neuroscientists a principled basis for choosing sound–color mappings.

1 Introduction

Few questions in the science of perception have attracted such sustained intellectual commitment - and produced such strong disagreement - as the relationship between musical pitch and color. Isaac Newton divided the visible spectrum into seven color bands and aligned them with the seven notes of the diatonic scale (Newton, 1704). Louis-Bertrand Castel built a keyboard instrument designed to project colors alongside musical tones (Castel, 1725). Alexander Scriabin notated a full color-organ part in his orchestral score *Prometheus: The Poem of Fire* (Scriabin, 1911). Olivier Messiaen described vivid, involuntary color experiences triggered by specific chord voicings (Messiaen, 1944). Today, smartphone applications map sound spectra to color in real time. Three centuries of effort by physicists, composers, instrument builders, synesthetes, and software engineers have all circled the same question - and arrived at irreconcilably different answers. Scriabin and Rimsky-Korsakov, contemporaries who both took the correspondence with utmost seriousness, agreed on little beyond assigning warm colors to C (Galeyev, 2007).

The disagreement is not for lack of musical reasoning - if anything, each proposal is over-determined by it. Newton's seven spectral bands were not chosen by eye: he deliberately partitioned the spectrum to echo the seven unequal steps of the Dorian diatonic scale, importing the ratios of the musical octave into the geometry of the prism (Newton, 1704; Gage, 1999). Castel's *clavecin oculaire* literalized the analogy into a machine, pairing each semitone with a pigment so that a harpsichordist could "paint" chords as they were played (Castel, 1725; Peacock, 1988). A century later the color-organ tradition - Rimington's electric *Colour-Organ* foremost among them - rebuilt the same dream with arc lamps and gelatin filters, mapping the chromatic scale onto a calibrated hue wheel (Rimington, 1912; Peacock, 1988). The composers who cared most disagreed most instructively: Scriabin derived his *luce* part in *Prometheus* from the circle of fifths, so that keys a fifth apart received neighboring hues, whereas Rimsky-Korsakov reported direct chromesthetic sensations that followed no such rule, and Messiaen described seeing shifting complexes of color bound not to single pitches but to specific chord voicings and modes (Scriabin, 1911; Messiaen, 1944; Galeyev, 2007). Older still, the Chinese *Wuxing* tradition linked the five pentatonic tones to the five phases and their canonical colors, embedding pitch–color correspondence inside a cosmological rather than perceptual logic (Gage, 1999). Each tradition is internally coherent; collectively they are mutually contradictory.

This persistent disagreement is worth taking seriously as evidence. Instead of asking which mapping is correct, we ask whether there is measurable structure in three centuries of pitch–hue proposals, and if so, which theoretical account best explains it. To do this we treat the historical corpus itself as a dataset-something that, to our knowledge, no prior survey has done quantitatively-and subject it to statistical analysis.

The stakes extend well beyond historical curiosity. A principled, data-driven account of pitch–hue correspondence would unlock concrete applications across several fields. In *music education*, color coding is already widely used to teach pitch, notation, and harmony to beginners (Boomwhackers, Figurenotes, and the colored-bell Suzuki and Kodály materials all assign colors to notes); knowing which mappings are perceptually and affectively natural rather than arbitrary would let educators design materials that learners find intuitive rather than confusing. In *accessibility and sensory substitution*, robust audiovisual mappings underpin tools that render music visible for the deaf and hard-of-hearing and sound audible-as-color for assistive displays. In *human–computer interaction and generative media*, real-time audio-to-color engines drive music visualizers, lighting design, and the rapidly growing class of AI audiovisual generators, all of which currently rely on ad hoc or designer-chosen palettes (Baione et al., 2026). And in *cognitive neuroscience*, a validated congruency criterion is a prerequisite for any experiment that hopes to localize the neural substrate of cross-modal binding - without one, “congruent” and “incongruent” stimuli are defined circularly by the very intuitions under study. Resolving the structure of pitch–hue correspondence is therefore not an end in itself but an enabling step for pedagogy, assistive technology, creative tooling, and perceptual neuroscience alike.

Cross-modal correspondences (CMCs) are systematic matching tendencies between perceptual dimensions from different sensory domains, observed reliably across individuals in the general population (Spence, 2011). Pitch–hue is among the most studied yet least settled CMC pairings.

Behavioral evidence is frustratingly weak: Ward, Huckstep, and Tsakanikos (2006) found that participants associate higher pitches with lighter, more yellow-shifted colors, but the effect is small, highly variable, and entangled with brightness and saturation confounds (Ward et al., 2006). Rather than adding yet another explicit matching experiment to an already crowded literature, we exploit a richer and largely untapped empirical resource: three centuries of independent pitch–hue proposals produced by agents as diverse as Pythagorean cosmologists, chromesthetic synesthetes, music educators, and digital application designers. Any signal these proposals carry should show up in the structure of their agreement and disagreement, which we now have the tools to measure.

Four accounts have been proposed to explain CMCs in general and pitch–hue correspondences in particular (Spence, 2011; Deroy et al., 2016). The **structural account** (Marks, 1978) holds that correspondences arise from shared physical or mathematical properties: both pitch and hue are circular dimensions with interval-ratio relationships, and the pitch-class circle and the color wheel may share the same underlying geometry. The **semantic account** (Martino and Marks, 2001; Walker, 2016) holds that shared linguistic labels mediate correspondences: "C major" and "red" are associated because Western cultural history has described both with overlapping emotional vocabulary ("warm", "powerful", "grounded"). The **statistical account** (Parise and Spence, 2014) holds that correspondences arise from environmental co-occurrence regularities - high-pitched bird calls in bright morning light, low-frequency rumbles in dark overcast conditions - internalized through Bayesian learning. The **hedonic account** (Palmer et al., 2013) holds that stimuli evoking similar emotional responses are associated: correspondence operates through a shared affective dimension onto which both modalities project, rather than through direct physical or linguistic links.

The present study tests all four accounts against a newly constructed catalogue of 70 pitch–hue mapping systems, the largest assembled to date. The approach differs from prior work in three ways. First, instead of adopting any single reference map as ground truth, we quantify the *geometry of agreement* across systems, asking whether musical interval structure is preserved in color distance structure; this reference-free framing sidesteps the circularity of earlier comparisons. Second, we organize systems into eight theoretically motivated *reasoning families*-classified from source texts before any agreement is computed-and test whether a shared reasoning tradition predicts shared color assignment. This lets us test whether the global disagreement seen in prior surveys conceals coherent within-tradition structure-a form of meta-analytic heterogeneity in which pooling masks subgroup agreement. Third, we probe the semantic account directly using pretrained sentence-transformer models (Wang et al., 2020; Song et al., 2020; Nussbaum et al., 2024) trained on large natural-language corpora, turning the linguistic mediation hypothesis into a directly testable claim.

In this paper, we present a curated catalogue, a reference-free agreement framework, and a head-to-head comparison of theoretical accounts - quantitative targets for the behavioral and neural experiments that should follow.

2 Materials: Constructing the Historical Catalogue

2.1 Source Identification and Coverage

Primary sources were identified through three channels. First, four academic surveys of the sound–color relationship (Jewanski, 2009; Gage, 1999; Peacock, 1988; Wells, 1980) were reviewed systematically to extract every referenced pitch–color mapping system. Second, targeted searches in music history archives, color science literature, and instrument design records extended coverage to include practitioner systems - color organs, music education curricula, digital applications - underrepresented in academic surveys. Third, a bibliographic snowball procedure followed citations until no new systems emerged. The procedure yielded 70 candidate systems spanning from Chinese Wuxing theory (4th century BCE) to contemporary smartphone color–music applications.

2.2 Color Standardization

Historical color specifications are heterogeneous: Newton specifies spectral band positions; Scriabin specifies color names in Russian; Rimington specifies 19th-century pigment trade names; digital applications specify hexadecimal RGB codes; synesthetes report colors in free text. Each specification was classified into one of four source types: (1) spectral wavelength, (2) named pigment or dye, (3) natural-language description, or (4) modern color code (RGB/hex), and converted to CIELAB coordinates via a principled pipeline. Spectral wavelengths were converted to CIEXYZ using the CIE 1931 standard observer, then to CIELAB. Named pigments and dyes were matched to Munsell (Newhall et al., 1943) or Natural Color System (Hård and Sivik, 1981) references using published pigment databases. Natural-language descriptions were resolved by averaging the CIELAB coordinates of all XKCD Color Survey (Munroe, 2010) colors judged semantically compatible with the description. Modern color codes were converted via the standard sRGB-to-CIELAB transformation with D65 illuminant.

2.3 Inclusion Criteria

Each candidate system was evaluated against four criteria: (1) *Discrete pitch-class-to-color mapping*: the system must assign colors to discrete pitch classes. Systems mapping intervals, modes, timbres, keys that cannot be normalized to the 12 pitch classes, or continuous frequency were excluded, as were systems specifying only lightness or brightness without hue. Partial coverage (pentatonic, diatonic, or heptatonic systems) was retained; per-pitch statistics are computed over the systems that specify that pitch class, and interpolation of missing pitch classes was rejected because any interpolation scheme imposes structural assumptions that the analysis is designed to test. (2) *Convertible color specifications*: all assignments must be convertible to CIELAB coordinates. (3) *Traceable provenance*: each system must be attributable to a published source with sufficient detail to verify color assignments. (4) *Independence*: directly derived or copied systems were merged with their originals.

After applying these criteria, 25 of the candidate systems identified through the survey channels were retained for the primary quantitative analyses; 14 of the retained systems specify all 12 chromatic pitch classes. The remaining catalogued systems are documented but excluded from agreement computations. Supplementary Table S1 lists all 70 systems with year, source, family, specification type, retention status, and exclusion reason. Six further systems meeting the same criteria entered the catalogue through the snowball procedure after the primary analyses were run, bringing the total of retained systems in the final catalogue to 31. For two of them, Castel's *clavecin oculaire* (1725) and Rimington's color-organ assignments (1895)-both discussed, but not fully tabulated, in prior surveys (Peacock, 1988; Gage, 1999)-complete 12-tone color specifications were assembled from the primary sources. The remaining four are Field's chromatic equivalents (Field, 1835), Wells' proportional frequency system (Wells, 1980), Pridmore's psychophysical transposition (Pridmore, 1992), and a contemporary smartphone Soundmap application. Unless noted otherwise, results below are reported for the 25-system primary set; a sensitivity analysis on all 31 retained systems (flat agreement 0.237, Rayleigh $R = .60$, semitone–hue correlation $r = .49$) leaves every conclusion unchanged.

2.4 Reasoning-Family Classification

The 70 systems were classified into eight reasoning families (F1–F8) based on the design rationale of each system *before* agreement scores were computed, to avoid circular analysis. Two independent coders assigned each system to a family from the original source text; disagreements ($\approx$12% of cases) were resolved through discussion. The eight families are: F1 Spectral Analogy (22 systems), F2 Circle of Fifths (4), F3 Cosmological (14), F4 Emotion-Mediated (3), F5 Synesthetic (7), F6 Esoteric (2), F7 Psychophysical (8), and F8 Pedagogical (4); the remaining six catalogue entries are meta-level sources (survey compilations and documented rejections of fixed mappings) that name no single design rationale and were left unclassified. Twelve systems invoke a secondary rationale in addition to their primary one; the priority rule described below assigns each to a single primary family.

The eight categories follow directly from the four theoretical accounts the study sets out to test, refined to the granularity actually present in the historical record. The *structural* account corresponds to F1 (Spectral Analogy, Newton's prism metaphor) and F2 (Circle of Fifths, harmonic interval geometry); the *hedonic* account to F4 (Emotion-Mediated) and, indirectly, F8 (Pedagogical), where designers optimize for what feels natural to a learner; the *semantic* and culturally transmitted accounts to F3 (Cosmological, pitch–color links inherited from metaphysical systems) and F6 (Esoteric, Theosophical and occult schemes). F5 (Synesthetic) isolates first-person chromesthetic report, and F7 (Psychophysical) isolates systems derived from explicit perceptual measurement. Each family is defined by a single text-based inclusion criterion-the *stated* design rationale of the system, not its resulting colors (Table 1). Coders applied the criteria in fixed order F1 through F8, assigning each system to the first family whose criterion it met; this priority rule resolves the small number of systems invoking more than one rationale (a color organ justified on both spectral and pedagogical grounds is coded F1, its primary stated basis). Eight is the smallest number of families that keeps the four accounts distinguishable while separating the two recurring expressions of structural and hedonic reasoning that the record contains; finer splits produced singleton families, and coarser ones merged rationales the sources treat as distinct. Retaining the eight-way scheme rather than collapsing to the four accounts lets a single account succeed in one guise and fail in another (structural reasoning expressed spectrally versus harmonically, for example)-a distinction that proves diagnostic.

2.5 Color Distance Metric

All inter-color distance computations used the CIEDE2000 formula (Sharma et al., 2005):

$$\Delta E_{00} = \sqrt{\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2 + R_T \frac{\Delta C'}{k_C S_C} \frac{\Delta H'}{k_H S_H}} \tag{1}$$

where $\Delta L'$, $\Delta C'$, $\Delta H'$ are lightness, chroma, and hue differences; S_L, S_C, S_H are empirically calibrated weighting functions; $k_L = k_C = k_H = 1$; and R_T corrects perceptual non-uniformities in the blue-violet region. CIEDE2000 was preferred over raw CIELAB Euclidean distance to

Table 1: Operational definitions of the eight reasoning families. Each system was assigned to the first family (in order F1–F8) whose stated design rationale it satisfied. Counts are catalogue-level (of the 70 candidate systems); the subset retained for agreement analysis is smaller (see Section 3 and Table 2).

ID	**Family**	**Defining criterion (stated rationale)**	**Representative system**
F1	Spectral Analogy	Color assigned by analogy to the optical spectrum or the rainbow (Newton's prism metaphor)	Newton (1704); Castel (1725)
F2	Circle of Fifths	Color assigned to preserve harmonic interval geometry, neighboring keys receiving neighboring hues	Scriabin (1911)
F3	Cosmological	Color inherited from a metaphysical or cosmological system (phases, planets, elements)	Chinese *Wuxing*
F4	Emotion-Mediated	Color chosen to match the explicitly named emotional character of the pitch or key	Mood-based key–color charts
F5	Synesthetic	Color reported as a first-person, involuntary chromesthetic experience	Rimsky-Korsakov; Messiaen (1944)
F6	Esoteric	Color derived from Theosophical, occult, or spiritual doctrine	Theosophical scales
F7	Psychophysical	Color derived from explicit perceptual or psychophysical measurement	Pridmore (1992)
F8	Pedagogical	Color chosen to be intuitive for teaching pitch or notation to learners	Boomwhackers; colored-bell methods

ensure that differences in the blue–green region are not artifactually underweighted. An **agreement score** was derived as the inverse of normalized mean pairwise CIEDE2000 distance:

$$\text{agreement} = \frac{1}{1 + \overline{\Delta E_{00}}/12} \tag{2}$$

The denominator of 12 was calibrated so that a score of 0.60 corresponds to a mean CIEDE2000 distance of ≈8 units, a conventional benchmark above which color pairs are judged clearly distinct under standard viewing conditions (Sharma et al., 2005); we treat 0.60 as an explicitly chosen benchmark for "strong consensus" rather than as a perceptual constant. A chance baseline for the agreement score was estimated by passing simulated systems of 12 colors, drawn uniformly from the sRGB gamut, through the identical pipeline (2,000 simulated ensembles per corpus size). For ensembles of 24 random systems the simulated chance agreement is 0.212 (95% interval $[0.196, 0.231]$); for ensembles of 15 it is 0.212 $[0.192, 0.240]$, with wider intervals at smaller ensemble sizes (full grid in Supplementary Table S2). Observed scores are compared against this simulated distribution wherever "chance" is invoked below.

One implementation detail matters for the hue-angle analyses reported below. Consensus hue angles were computed by averaging the CIELAB a^*/b^* coordinates across systems and taking the hue angle of the mean vector. This is equivalent to a chroma-weighted circular mean: it prevents instability near the achromatic axis by construction, since near-achromatic assignments (e.g., Scriabin's "steely" grays) contribute vectors of small magnitude and thus receive correspondingly little weight.

3 Analyses and Results

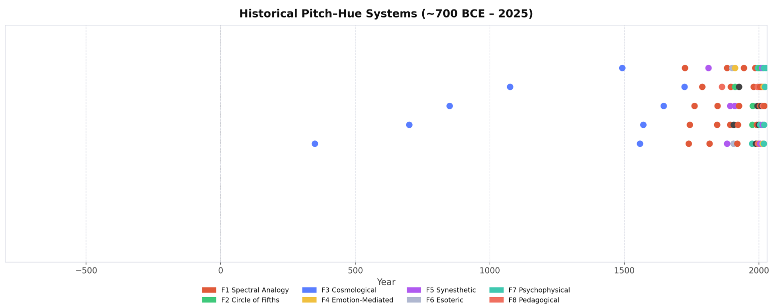
The four theoretical accounts are taken up in turn, each method stated immediately before the result it yields. The scoreboard, in brief: the hedonic account performs best (Procrustes disparity 0.559 in affective space); the structural account earns partial credit (interval ordering, but not specific assignment); the statistical account passes its distinctive variance prediction; and the semantic account contributes a channel that cannot be fully separated from affect. Four claims were designated as primary-family-level heterogeneity, the EPA Procrustes alignment, the interval–variance correlation, and the timbral embedding alignment-and multiplicity across the full test battery is addressed in the Discussion and Supplementary Table S7. All agreement computations rest on the CIEDE2000 distance and agreement score defined in Equations 1 and 2.

3.1 Three Centuries, Seventy Systems

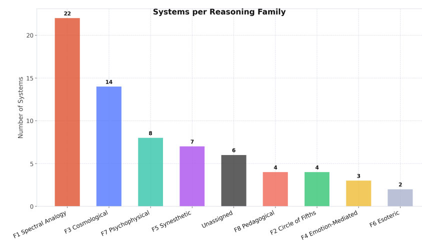
The 70 catalogued systems span 23 centuries of intellectual history; the retained corpus is concentrated in the three centuries from Diez (1723) and Castel (1725) to the present-the “three centuries” of our title-with three ancient cosmological systems (Chinese *Wuxing*, Ancient Persian, Indian svara) as the only earlier retained entries. The temporal distribution is itself revealing. Figure 1 shows that proposal density peaks sharply in the 19th and early 20th centuries - the era of color organs and synesthetic aesthetics - and surges again in the digital age, with a pronounced trough from approximately 1930 to 1980 when serialist and atonal movements displaced color from the mainstream musical imagination. Newton’s shadow is long: the Spectral Analogy family (F1) dominates the corpus (Figure 1, right), with his prism metaphor driving more than a third of all proposals across three centuries.

Of the 70 candidate systems, 31 met all four inclusion criteria (25 in the primary set); the dominant sources of exclusion were mappings defined over intervals, modes, or timbres rather than pitch classes, and specifications restricted to lightness without hue. That Newton’s own 7-note scheme left the five chromatic pitch classes unassigned is a choice that cascaded through centuries of followers.

As a check on the hand-coded taxonomy, we clustered the systems of the merged family corpus (Section 3) by their pairwise CIEDE2000 distance profiles using Ward-linkage hierarchical clustering (Ward, 1963). The resulting partition recovers the hand-coded families only weakly (adjusted Rand index $= .23$ at $k = 5$, $.12$ at $k = 8$). This is informative rather than disqualifying: the families were coded from *stated design rationale*, not from the colors themselves, and the low Rand index confirms that the two codings are not redundant-systems within a reasoning tradition need not produce similar palettes, which is precisely the question the family-level agreement analysis quantifies.



(a) Historical timeline of pitch–hue mapping proposals (Newton 1704 to present). Proposal density peaks in the 19th–early 20th century and again in the digital age.



(b) Distribution of the 70 systems across the eight reasoning families. F1 (Spectral Analogy) is the largest.

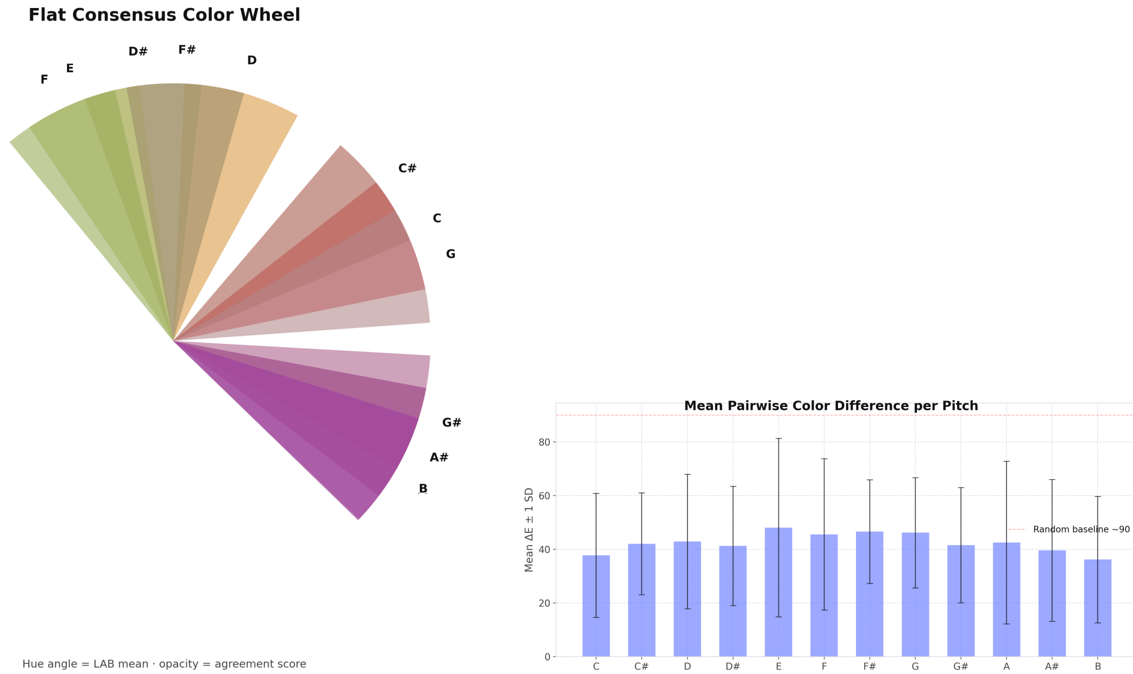
Figure 1: Overview of the historical catalogue. Left: timeline. Right: family distribution.

3.2 Global Consensus Is Weak but Non-Random

The global picture is clear: there is no universal pitch–hue mapping in this corpus. CIEDE2000 agreement across all 25 retained systems ranged from 0.20 to 0.30 across the 12 pitch classes (mean 0.221), well below the 0.60 benchmark for strong consensus and close to the simulated chance level of 0.212 for random color systems (Section 2.5); only G♯ (0.30) exceeds the upper bound of the simulated chance interval. Every pitch class spans a broad arc of hue space when all traditions are pooled (Figure 2, left), so the data do not support any claim that a particular system captures the "true" pitch–hue correspondence.

The assignments are nonetheless far from random. The Rayleigh test, applied to the 12 consensus hue angles (chroma-weighted circular means across the 25 retained systems), returned $R = 0.66$; a Monte Carlo test against 12 hues drawn uniformly on the circle gives $p = .003$. The mean resultant direction is $24°$, in the orange–red region: a clear warm-color bias. The pattern of variance is also informative (Figure 2, right): pitch classes C and G, the tonal anchors of Western harmony, show the lowest inter-system spread, while the chromatic alterations (C♯, D♯, F♯, G♯, A♯)

show the highest-a trace of Western tonal hierarchy in a corpus dominated by Western sources.



(a) Global consensus wheel. Each wedge is positioned at the chroma-weighted circular-mean hue for one pitch class, and wedge opacity encodes the agreement score. The uniformly low opacities and the wide dispersion of mean hues reflect the absence of a reliable global consensus.

(b) Per-pitch-class CIEDE2000 standard deviation across all 25 retained systems. C and G show the lowest variance (tonal anchors); chromatic alterations show the highest variance.

Figure 2: Global consensus analysis. Pooled agreement is low, but hue assignments are non-uniformly distributed (Rayleigh $R = 0.66$, Monte Carlo $p = .003$), indicating that the mapping is structured but tradition-specific rather than universal.

3.3 The Eight Families: Structure Within Traditions

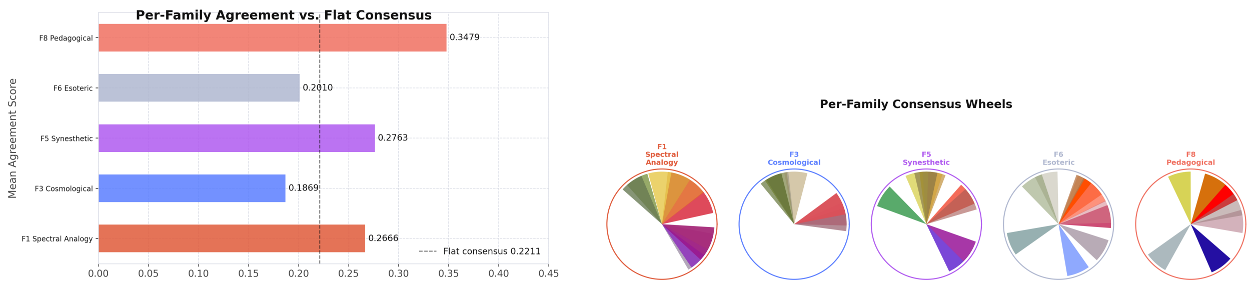
The family-level analysis is where the paper's central descriptive claim is established. This analysis uses a *merged family corpus*: the retained catalogue systems plus independently digitized versions of several historical systems from our initial source collection, giving 39 systems across the five families with at least two members (Table 2). Three systems in F1 appear twice (Castel, Field, and Rimington, each digitized from two editions); deduplicating them changes the F1 score by less than 0.001, so the merged counts are reported as analyzed. Breaking the corpus into reasoning families reveals a consistent ordering that the flat analysis hides. The Pedagogical family (F8) leads at 0.348, 57% above the global consensus. Synesthetic systems (F5) follow at 0.276, and Spectral Analogy (F1) at 0.267. The Cosmological family (F3) goes the other way: it falls *below* the flat baseline at 0.187 vs. 0.221.

To quantify the uncertainty of these small-sample scores, each family estimate is accompanied by a bootstrap confidence interval (systems resampled within family, 10,000 draws, resamples constrained to contain at least two distinct systems), and the within-family advantage was tested by a family-label permutation test in which the family labels were shuffled across the merged corpus and the family agreement scores recomputed (10,000 permutations). The outcome is sobering and worth stating plainly: no single family's deviation from the pooled baseline is individually significant (F1 $p = .10$, F8 $p = .33$, F5 $p = .40$; F3's below-baseline score has $p = .91$ against the greater-than null, i.e., it sits in the lower tail as predicted), and a global heterogeneity permutation test-the label-shuffle null for the variance of family means-is likewise non-significant ($p = .81$). With 3–5 systems in four of the five families, the family ordering reported here is a descriptive, effect-size-level pattern whose confirmation requires an expanded corpus. Interval estimates and permutation p-values are reported in Supplementary Table S2.

Table 2: Within-family versus flat-catalogue CIEDE2000 agreement scores in the merged family corpus; N is the number of systems contributing to each family's score. Within-family agreement is reported for families with $N \geq 2$ systems; F2 (Circle of Fifths), F4 (Emotion-Mediated), and F7 (Psychophysical) each contributed a single system and are omitted. F8, F5, and F1 exceed the flat baseline; F6 and F3 fall below it. The flat baseline is computed on the 25-system primary set. Bootstrap intervals and permutation p-values are given in Supplementary Table S2.

Family	**Tradition**	N	**Agreement**	**Δ vs. flat**
F8	Pedagogical	3	**0.348**	$+0.127$
F5	Synesthetic	5	0.276	$+0.055$
F1	Spectral Analogy	24	0.267	$+0.046$
F6	Esoteric	3	0.201	-0.020
F3	Cosmological	4	0.187	-0.034
Flat (all retained systems)		25	0.221	–

Figure 3 illustrates these results visually. The bar chart (left) plots within-family agreement against the flat baseline; the wheel panel (right) shows the mean hue assignment for each pitch class within each tradition - confirming that these are not minor quantitative differences but categorically distinct color philosophies.



(a) Within-family agreement scores compared against the flat baseline (dashed line). F8 and F5 substantially exceed global agreement; F3 falls below it.

(b) Per-family consensus wheels. Point size is proportional to within-family agreement. The traditions assign color through fundamentally different color philosophies.

Figure 3: Family-level analysis. Within-family structure substantially exceeds global agreement for Pedagogical and Synesthetic traditions, while Cosmological systems fall below the flat baseline.

The Cosmological family's result deserves emphasis: it is the most important negative finding in the dataset. Centuries of cosmological theorizing produce *more* disagreement than pooling all retained systems together. The four analyzed F3 systems span genuinely distinct cosmological traditions-the Chinese *Wuxing* five-phase system, the Ancient Persian system, the Indian svara associations, and Diez's planetary note–color scheme (1723)-so the low within-family agreement reflects divergence between independent metaphysical frameworks, not noise within one. This argues against any account that treats cosmological reasoning as a primary driver of pitch–hue correspondences.

The Pedagogical family points the other way, but two alternative explanations must be addressed before reading its convergence as affective. The first is lineage: commercial teaching materials copy one another, so inclusion criterion 4 (independence) was applied strictly to F8, merging derivative products with their sources. The three analyzed pedagogical systems-Sudre's Solresol color code (France, 1862), Boomwhackers (United States, 1995), and Figurenotes (Finland, 1996)-arose in different countries, markets, and eras with no documented lineage among them. The second is structure: educators might simply reuse rainbow order, which would make the convergence spectral rather than hedonic. We therefore compare the F8 consensus mapping against the canonical spectral-order mapping in Supplementary Analysis S3; the two are close (mean angular distance $35°$ after optimal rotation), so the pedagogical advantage should be read as convergence on a spectrally anchored, affectively comfortable mapping rather than as purely affective agreement. With these caveats stated, the convergence remains striking: educators across eras, optimizing for what feels natural to untrained learners, agree more than any other tradition. Figure **??** shows the complete mapping panel organized by family.

The complete per-system mapping panel, organised by reasoning family, is shown in Supplementary Figure S15. The traditions also differ in *how* they encode harmonic relationships (Supplementary Figure S10). The Spectral Analogy and Synesthetic families embed substantially stronger interval-to-hue structure ($\rho \approx 0.6$) than the Esoteric family, and all three assign more similar hues to consonant than to dissonant intervals - an early hint of the structural signal we quantify next.

3.4 Testing the Structural Account: Does Interval Geometry Predict Hue Geometry?

Method. We tested six interval-to-hue hypotheses, ordered from the strongest and most specific claim to the weakest (Table 3). The interval–distance correlation (H2) is computed over the $\binom{12}{2} = 66$ pitch-class pairs of the pooled consensus. Two properties of this design constrain inference: circular pitch-class distance collapses the eleven interval sizes to six distinct values (1–6 semitones), and pairwise distances are not independent. Statistical significance was therefore assessed by a pitch-label permutation test in addition to parametric inference: the 12 pitch-class labels were randomly reassigned across color assignments 10,000 times to generate a null distribution of correlation values. The point-prediction hypotheses (H1, H3–H6) are evaluated descriptively, by comparing the observed mean hue distance for the relevant interval against the predicted angular range. We then evaluated six principled structural mapping theories (Table 4), each a complete first-principles recipe for assigning hues to pitches, on two complementary statistics. The first is a reference-free *geometry alignment* score: the Spearman correlation between the $\binom{12}{2}$ pairwise circular pitch-step distances and the corresponding pairwise hue angular distances of the theory's own predicted mapping,

$$\rho_{\text{geom}} = \text{Spearman}\left(\{d_{\text{pitch}}(i,j)\}_{i<j}\,,\left\{\left|\hat{h}_i - \hat{h}_j\right|_{\text{circ}}\right\}_{i<j}\right) \tag{3}$$

where $\hat{h}$ are the predicted hue angles and $|\cdot|_{\text{circ}}$ denotes circular (wrap-around) angular distance. This statistic asks whether the theory realizes an interval-to-arc isomorphism at all, without privileging any historical map as ground truth; its chance level was estimated from 10,000 random hue assignments (null mean $\rho_{\text{geom}} \approx 0$, 95th percentile 0.23). The second statistic is the *consensus angular error*: the mean circular distance between the theory's predicted hue and the observed pooled-consensus hue across the 12 pitch classes. Because two hue angles drawn independently at random differ by $90°$ on average, a theory unrelated to the data scores $\approx 90°$ on this measure (the *random-hue null*); exact Monte Carlo p-values are reported against that null. For calibration we also report two external benchmarks: a perfect spectral-order mapping ($\rho_{\text{geom}} = 1.00$) and the circle-of-fifths mapping ($\rho_{\text{geom}} = -.19$). That the spectral mapping attains the maximal geometry score is itself a finding: the pooled historical consensus preserves spectral hue order almost exactly, so the spectral mapping functions as the effective ceiling of the scale.

Results. The structural account earns a qualified pass. Of the six interval hypotheses, the weakest (H2, monotone ordering), the most local (H6, adjacent hues for semitone neighbors), and the broad-band intermediate prediction (H5) pass; every point prediction targeting a *specific* angular value-complementary or triadic-fails (Table 3). The positive correlation between semitone distance and mean hue distance ($r = .393$, parametric $p = .001$, pitch-label permutation $p = .002$) is clear: across all retained systems, larger musical intervals consistently map to larger color distances (Figure 4, left); per-interval values are in Supplementary Table S8.

The account's strongest prediction, however, fails. The claim that harmonically complementary intervals should map to visually complementary hues ($\approx$180° apart) is rejected (Figure 4, right): perfect fifths averaged only 80.7° of hue separation, less than half the predicted value. The interval–

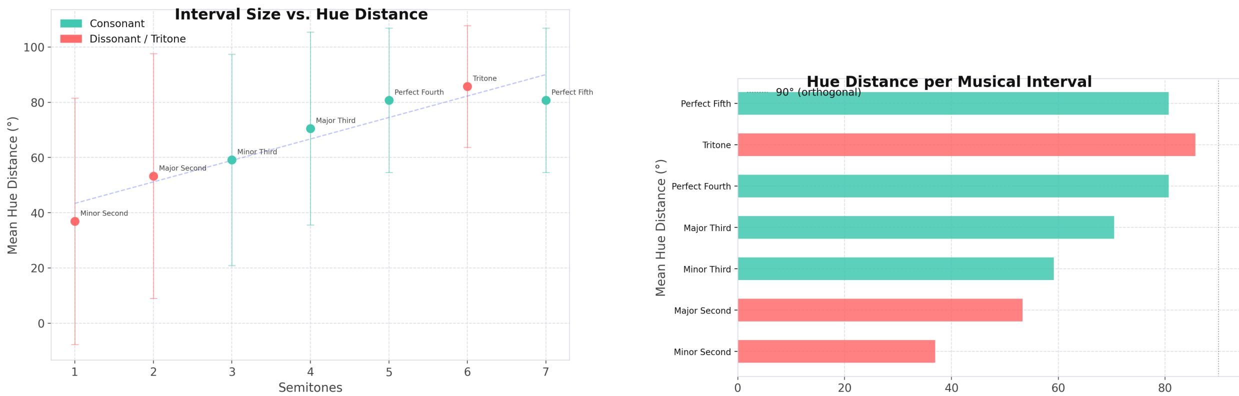
Table 3: Six structural hypotheses linking musical intervals to color-wheel geometry, with outcomes. H1–H6 span from the strongest claim (perfect fifth maps to complementary hue at 180°) to the weakest (monotone positive correlation between interval size and hue distance). Verdicts for the point predictions (H1, H3–H6) are descriptive comparisons of the observed mean hue distance (Table **??**) against the predicted range; only H2 receives formal (permutation) inference. Note that circular pitch-class distance makes the perfect fourth and perfect fifth equivalent (both distance 5), so H1 and H5 address the same observed value.

H	**Interval**	**Semitones**	**Prediction**	**Result**
H1	Perfect fifth	7	Complementary hues ($\sim$180°): most consonant interval $\leftrightarrow$ maximal color contrast	Fail (80.7°)
H2	All intervals	1–11	Semitone distance correlates positively with hue distance (interval-to-arc isomorphism)	**Pass** ($r = .393$, perm. $p = .002$)
H3	Tritone	6	Complementary hues ($\sim$180°): most dissonant interval $\leftrightarrow$ maximal color contrast	Fail (85.7°)
H4	Major third	4	Triadic hue separation (90–150°, centered on 120°): three major thirds tile the octave, mirroring three-fold color-wheel symmetry	Fail (70.5°)
H5	Perfect fourth	5	Intermediate hue distance (70–140°): between analogous and complementary	**Pass** (80.7°)
H6	Minor second	1	Adjacent hues ($<$60°): chromatic neighbors $\leftrightarrow$ analogous, closely spaced colors	**Pass** (36.9°)

Table 4: Six principled structural theories, evaluated on geometry alignment (Equation 3) and on angular error against the pooled consensus.

Abbr.	Name	Core principle
IRP	Interval-Ratio Preservation	Maps 12 pitch classes onto the color wheel so that semitone ratios mirror hue-angle ratios, optimizing rotation to maximize the consonance–color-distance correlation.
FRI	Frequency-Ratio Isomorphism	Maps C4–B4 on a log-frequency scale to the visible spectrum (380–780 nm), revising Newton's linear mapping for the logarithmic nature of pitch perception.
HEM	Harmonic Entropy Matching	Pairs pitches with low harmonic entropy (consonant, simple ratios) to focal colors (Berlin and Kay, 1969) on the grounds that shared perceptual salience mechanisms link consonance and color centrality.
MPA	Multidimensional Perceptual Alignment	Constrains pitch height (mel scale) to lightness and spectral centroid to color temperature; hue is solved as the residual dimension.
PDM	Perceptual Distinctiveness	Maps tonal hierarchy positions (probe-tone ratings; Krumhansl, 1990) to color naming consensus, pairing perceptually central pitches with the most-named colors.
SSC	Spectral Shape Correspondence	Aligns RDMs of piano harmonic spectra with hue spectral profiles, testing whether the physical frequency content of a tone predicts the hue whose spectral reflectance it most resembles.

distance relationship saturates near 85° and never approaches the complementary region. Interval structure can account for the ordinal pattern, then, but not for the specific mapping.

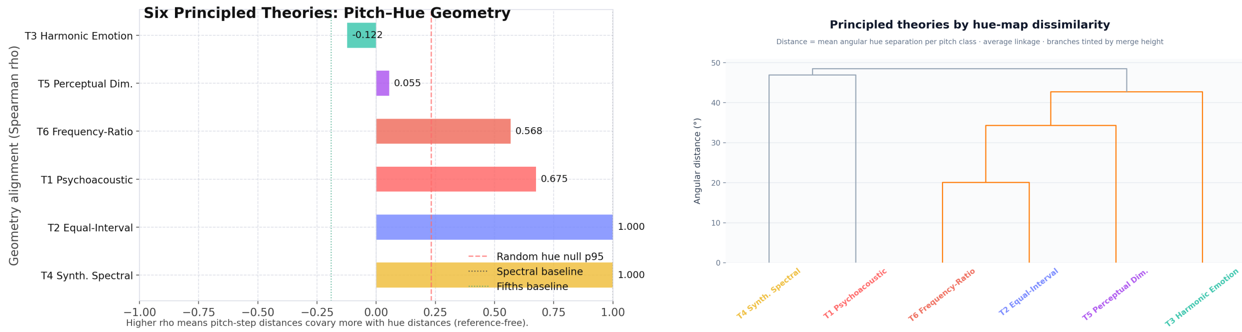


(a) Semitone distance vs. mean hue distance across all historical systems ($r = .393$, $p = .002$). The positive trend indicates partial support for the structural account.

(b) Mean hue distance by interval type. The dashed line at 180° marks the complementary-color prediction for the tritone and perfect fifth. Neither interval approaches this value.

Figure 4: Structural account: interval distance predicts hue distance, but the strong complementary-hue prediction fails.

Among the six formalized structural theories, the two statistics dissociate informatively. On internal geometry alignment (Equation 3), four theories exceed the random-hue 95th percentile of $\rho_{\text{geom}} = .23$: IRP and SSC realize a perfect interval-to-arc isomorphism by construction ($\rho_{\text{geom}} = 1.00$, empirical $p < .001$), followed by MPA (.68, $p < .001$) and FRI (.57, $p < .001$); PDM (.06, $p = .28$) and HEM ($-.12$, $p = .85$) fail. On angular error against the pooled historical consensus, however, only IRP (46.2°, Monte Carlo $p = .001$) and FRI (54.9°, $p = .009$) score significantly below the 90° random-hue null; HEM (70.7°, $p = .10$), MPA (75.8°), PDM (76.3°), and SSC (81.2°) do not (Figure 5, left). The pattern is coherent: theories that both encode the isomorphism *and* anchor it to the spectrum (IRP, FRI) match the historical record, whereas SSC shows that an isomorphism anchored elsewhere lands far from the consensus, and theories routing through intermediate perceptual dimensions (PDM, HEM) capture neither. Notably, IRP and FRI-derived independently, through different mathematical formalisms-cluster together in the theory dendrogram (Figure 5, right; pairwise distance 40.1°). Two independent theories landing on the same below-chance geometry is good evidence that a structural isomorphism signal is present in the historical corpus, even if it is a partial one.



(a) Consensus angular error for the six structural theories. Only IRP and FRI score significantly below the random-hue null (∼90°); HEM, MPA, PDM, and SSC do not. Vertical lines mark spectral/fifths benchmarks and the random-hue null.

(b) Dendrogram of the six theories by mutual geometry distance. IRP and FRI form a tight cluster at low linkage distance, confirming their structural kinship.

Figure 5: Structural theory evaluation. Only direct isomorphism theories (IRP, FRI) achieve below-chance geometry error.

The pairwise theory alignment heatmap and predicted hue wheels for all six theories are provided in Supplementary Figures S11–S12; IRP is the closest match to the historical consensus, though it diverges substantially in the middle register (D–G♯)- showing that even the best structural theory captures ordering without pinning down assignment.

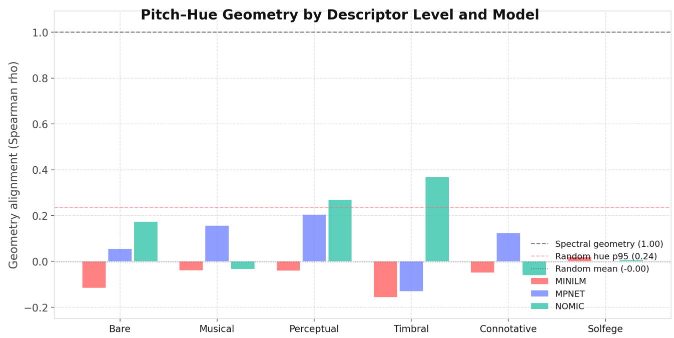
3.5 Testing the Semantic Account: Is the Correspondence Carried by Language?

Method. We probed three pretrained sentence-transformer models - MiniLM (Wang et al., 2020), MPNet (Song et al., 2020), and Nomic Embed (Nussbaum et al., 2024) - to ask whether the linguistic descriptions of pitches and colors carry pitch–hue structure on their own. Pitch classes (C through B) were embedded using six descriptor levels: bare name, musical context (scale degree, key relationships), perceptual (frequency, register), timbral (instrument descriptions), connotative (emotional/associative character), and solfège syllable. Two independent descriptor sets were used: a primary *LLM-generated* set produced with an explicit constraint against color, brightness, and temperature words (enforced by an automated validator against a 35-term blocklist), so that any recovered geometry could not be a trivial artifact of color words leaking into the pitch descriptions, and a secondary *factual* set derived deterministically from music-theory constants (frequency, scale degree, interval structure, instrument registers), used as a robustness check on descriptor wording. Color-side embeddings used names from the XKCD Color Survey (Munroe, 2010) - 954 crowd-sourced color names with sRGB hex codes. For each pitch class, the top-k nearest XKCD colors (by cosine similarity) formed a centroid hue angle, and the resulting 12-point pitch–hue mapping was scored by geometry alignment: the Spearman ρ between all $\binom{12}{2} = 66$ pairwise circular pitch-step distances and the corresponding pairwise hue angular distances.

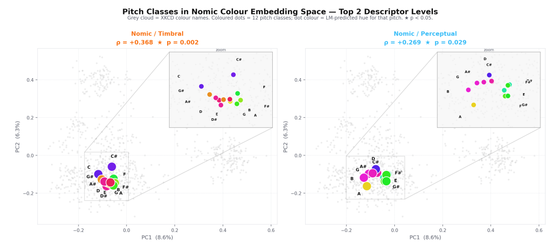
Results. The sentence-transformer probing yields a clear and interpretable result (Table 5). The strongest geometry signal is Nomic with timbral descriptors ($\rho = .368, p = .007$): statistically significant, well above the random-hue null, and consistent across permutation iterations. So the semantic account does capture something, but only partly-the timbral signal remains well below the spectral geometry baseline ($\rho = 1.000$). One caveat governs the interpretation throughout: timbral vocabulary is heavily affective, so this signal cannot be read as purely linguistic. We treat it as a semantics–affect entangled channel and return to the distinction in the Discussion.

The descriptor-level pattern points to the mechanism (Figure 6, left). Timbral and perceptual descriptors outperform bare names and solfège across all three models. Timbral language-"warm", "round", "bright", "cutting"-carries affective content that links pitch and color through shared experiential vocabulary, and the contrast with the failed solfège condition (MiniLM: $\rho = .019$, $p = .344$) makes pitch-name familiarity an unlikely confound. The PCA projection (Figure 6, right) shows how: Nomic's timbral pitch embeddings separate lower from higher pitches along a dimension encoding "warm/round" versus "bright/cutting", which is the geometric substrate of the Spearman alignment. In short, the embedding space encodes that high notes are described as bright and that bright things are described as blue.

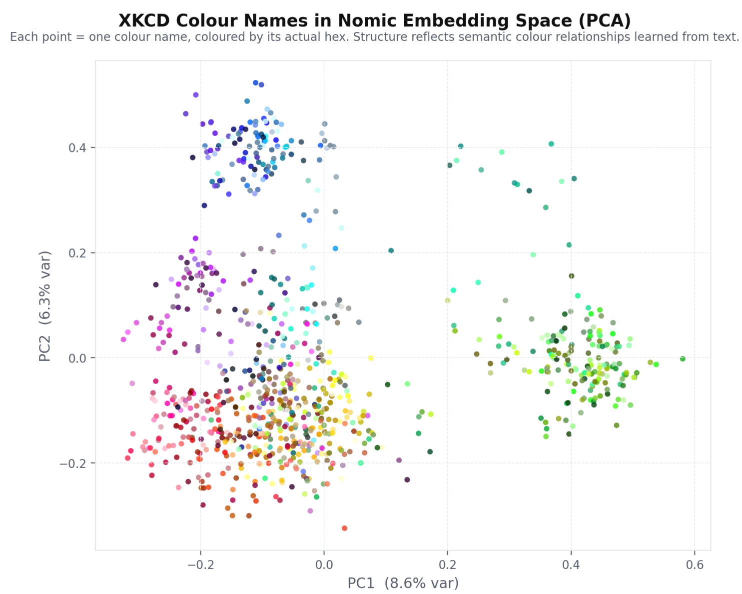
Results with the factual descriptor set are only partially consistent, and we report them as a genuine constraint on interpretation. Several factual conditions reach significance-MPNet with musical-context descriptors is the strongest ($\rho = .334$, permutation $p = .012$), and MiniLM



(a) Geometry alignment by descriptor level and model (LLM-generated descriptors). Nomic shows the strongest pitch–hue geometry under timbral/perceptual descriptor conditions.



(b) PCA projection of Nomic pitch-descriptor embeddings (timbral level). PC1 separates lower from higher pitches; PC2 encodes "warm/round" vs. "bright/cutting" timbral quality.



(c) Companion PCA projection of the 954 XKCD color-name embeddings (Nomic), each point colored by its actual hex value. Hue families form coherent clusters, showing that the color-side space against which pitch embeddings are matched itself encodes hue structure learned purely from text.

Figure 6: Semantic account: sentence-transformer probing reveals significant pitch–hue geometry under timbral descriptor conditions, consistent with a semantics–affect entangled channel.

Table 5: Sentence-transformer probing results. Geometry alignment is Spearman ρ between pairwise circular pitch-step distances and pairwise hue angular distances; p-values are from 10,000 label-shuffle iterations, computed against condition-specific nulls (the random-hue row is a global geometric reference, not the null used for the p-values). The top conditions of the 3 (model) $\times$ 6 (descriptor) grid are shown; the full 18-cell grid, with Benjamini–Hochberg corrected q-values, is given in Supplementary Table S4.

Model	**Descriptor level**	**Geometry ρ**	**Perm. p**
Nomic	Timbral	**0.368**	.007
Nomic	Perceptual	0.269	.044
MPNet	Perceptual	0.203	.048
Nomic	Bare names	0.173	.070
MPNet	Musical context	0.156	.060
MPNet	Connotative	0.124	.124
MPNet	Bare names	0.055	.226
MiniLM	Solfège	0.019	.344
Spectral geometry baseline		1.000	–
Circle-of-fifths baseline		-0.191	–
Random-hue null (mean / 95th pct.)		-0.001 / 0.235	–

($\rho = .265, p = .048$) and MPNet ($\rho = .197, p = .034$) recover geometry under timbral descriptors-but the model–level profile shifts: Nomic's timbral signal, the strongest cell of the primary grid, is absent under factual wording ($\rho = -.01$). The full factual grid is given in Supplementary Table S4. Geometry alignment is therefore recoverable from more than one descriptor family, but which model–level cell carries it depends on wording-a sensitivity consistent with the entangled semantics–affect reading rather than with a wording-independent property of the embedding spaces. Per-model predicted hue wheels are shown in Supplementary Figure S13. Two controls are reported in Supplementary Figure S14: pitch–hue geometry remains weak across naming conventions (arguing against name-specific artifacts), and the emotion-mediation correlation ($\rho = .139$, $p < .001$) is consistent with the hedonic mediation examined next.

3.6 Corpus-Level Regularities: Testing the Statistical Account and Its Neighbors

Method. Ten corpus-level hypotheses were evaluated on the pooled consensus and per-pitch statistics of the 25 retained systems (Table **??**). Four operationalize the environmental co-occurrence account proper (H_{b4}, H_{b6}, H_{b7}, H_{b9}); the remainder test neighboring accounts and are attributed accordingly in the table: H_{b1}, H_{b3}, and H_{b8} probe structural predictions, H_{b2} probes a salience/affect asymmetry, H_{b5} and H_{b10} probe notation-driven and tradition-driven cultural variation. The Rayleigh test (Mardia, 1972) was used for the non-uniformity prediction (H_{b4}); because H_{b4} restates the non-uniformity already reported in Section 3.2, it is counted once in the overall tally. Correlational hypotheses use Pearson tests over the 12 pitch classes or the interval table; group-difference hypotheses use Welch t-tests.

Results. The battery produces three passes, six failures, and one inconclusive result (Supplementary Table S9). The clearest result is H_{b6}: the correlation between interval size and between-system variance is $r = -.94$, $p = .001$-though, because circular pitch-class distance takes only six distinct values, this correlation is computed over seven interval rows and its precision should be read accordingly. Historical mappers agree more about large intervals and disagree more about semitones, the pattern one would expect from statistical learning of a noisy environmental signal that constrains global ordering but not fine-grained assignment. H_{b4} adds support: hue assignments cluster toward warm colors, consistent with the environmental co-occurrence of low-frequency sounds and warm atmospheric conditions. H_{b7}, by contrast, is downgraded from the "pass" a naive reading would give it: the fifth–tritone difference is 5.0° against standard deviations of 22–26°, was not formally tested, and the fifth is circularly equivalent to the fourth; we report it as directionally consistent but inconclusive.

Two failures deserve comment. The failure of H_{b1} is the more surprising: pitch–lightness is among the most robust cross-modal correspondences in the behavioral literature (Marks, 1978; Ward et al., 2006), yet across the historical consensus the correlation between chromatic pitch order and assigned lightness is directionally *negative* ($r = -.53$, $p = .076$). The historical record encodes hue structure, not the pitch–lightness gradient-consistent with our inclusion criteria, which excluded lightness-only systems, but also a caution against reading historical proposals as transcriptions of perceptual correspondence. The failure of H_{b10} ($p = .89$) shows that the perfect-fifth hue distance is statistically indistinguishable across reasoning traditions; combined with the family analysis, it suggests that traditions differ in *which* colors they assign more than in how they space harmonically related pitches.

3.7 Testing the Hedonic Account: Do Pitch and Color Share an Affective Geometry?

Method. The hedonic account predicts that the organization of auditory and visual dimensions in a general sensorimotor representation is captured by affective structure. To test this, the cross-modal dimensions in the CMC database of the following section were each coded on 11 sensorimotor–affective dimensions (visual, auditory, tactile, gustatory, olfactory, interoceptive, motor, valence, arousal, magnitude, spatial), an inventory modeled on the Lancaster Sensorimotor Norms (Lynott et al., 2020); because the Lancaster norms cover words rather than the technical dimension labels used here, the profiles were coded by the authors, a construct limitation weighed in the Discussion. The dimension profiles were reduced to three principal components (68.3% of variance). Ten dimensions central to the pitch–hue question-pitch height, loudness, tempo, hardness on the auditory–tactile side; lightness, brightness, saturation, hue warmth on the visual side; size and pleasantness as cross-modal anchors-were then assigned coordinates in Evaluation-Potency-Activity (EPA) affective space (Osgood et al., 1957), drawing on published semantic-differential and color-affect norms (Ou et al., 2004; Hevner, 1935; Wedin, 1972). Procrustes alignment (Gower, 1975) found the rotation, reflection, and uniform scaling minimizing squared distances between the sensorimotor PCA configuration and the EPA configuration of the same ten dimensions. The **Procrustes disparity** is

$$d^2 = 1 - \left(\sum_{k=1}^{p} \sigma_k \right)^2 \tag{4}$$

where $\sigma_1, \ldots, \sigma_p$ are the singular values of $\mathbf{X}^\top \mathbf{Y}$, with $\mathbf{X}$ and $\mathbf{Y}$ the centered, unit-scaled configuration matrices; a disparity of 0 denotes perfect alignment and 1 no alignment. Significance was assessed by permutation of the correspondence assignments before alignment (10,000 permutations); the exact p-value and the mean and standard deviation of the null disparity distribution are reported in Supplementary Table S5, along with dimension-wise alignment statistics.

Results. The hedonic account gives the best result of the four. The Procrustes disparity between the sensorimotor and EPA configurations is 0.559, significantly below the random-correspondence permutation null (null mean 0.833, SD 0.106; $p = .027$) and the best single-account result across the four theoretical tests, indicating that the space in which auditory and visual dimensions are organized shares an affective geometry. Dimension-wise correlations between matched coordinates after alignment identify Evaluation as the dominant channel ($r = .74$, $p = .014$), with Potency ($r = .47$, $p = .17$) and Activity ($r = .37$, $p = .29$) weaker and individually non-significant (Supplementary Table S5). Figure 7 shows the configuration.

Evaluation (E) is the dominant aligning dimension: the good–bad, pleasant–unpleasant axis orders auditory and visual dimensions in the same way, so that what aligns pitch-space with color-space most reliably is shared hedonic value. Activity and Potency contribute positively but weakly. Converging evidence at the item level comes from the embedding analysis of the previous section: proximity between pitch and color terms in embedding space tracks shared valence–arousal profiles ($\rho = .139$, $p < .001$; Figure **??**, right). This pattern is consistent with Palmer and colleagues' (Palmer et al., 2013) demonstration of emotion-mediated music–color associations, and locates the mediation on the Evaluation axis: what seems to matter is the shared pleasantness of sounds and colors more than their shared energy or power.

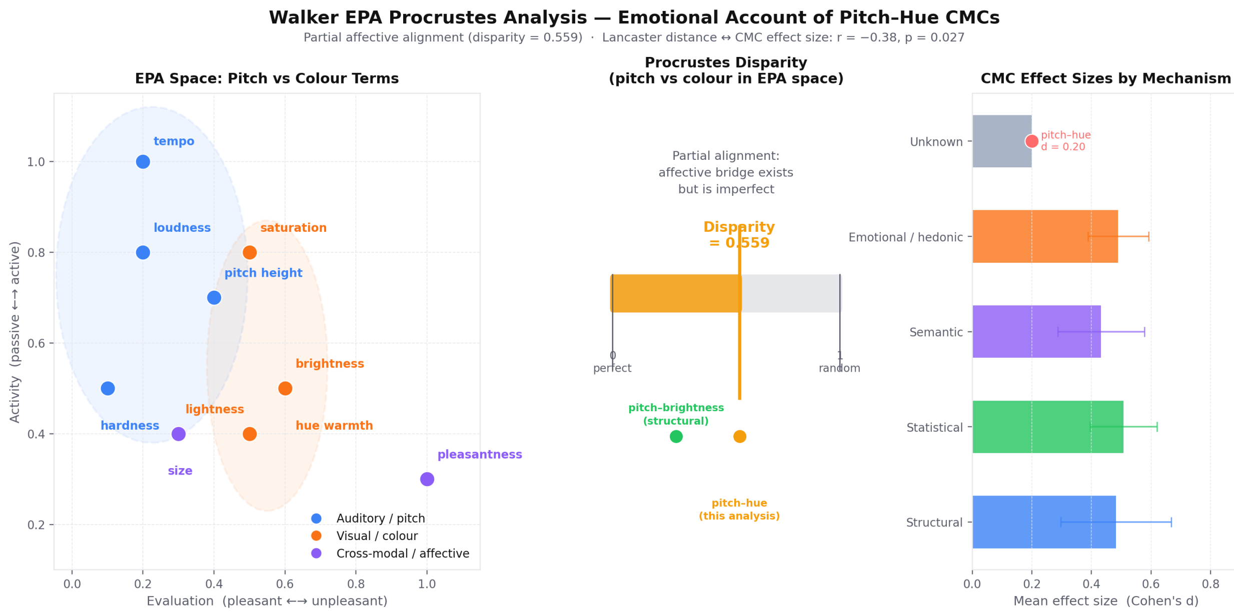


Figure 7: EPA Procrustes analysis. Left: the ten cross-modal dimensions positioned in affective space (Evaluation vs. Activity shown), with auditory dimensions (blue), visual dimensions (orange), and cross-modal anchors (purple); the auditory and visual clusters overlap along the Evaluation axis. Middle: the Procrustes disparity of 0.559 sits between perfect alignment (0) and the permutation null (0.833); the pitch–brightness correspondence, shown for calibration, aligns more closely than pitch–hue. Right: mean behavioral effect sizes of the catalogued CMC pairings grouped by proposed mechanism; pitch–hue ($d = 0.20$) is the weakest documented pairing.

3.8 Positioning Pitch–Hue in the CMC Landscape

Method. A structured database of 34 documented CMC pairings was compiled from the published literature (sources in Supplementary Table S6), each pairing coded for source and target modality, direction, mechanism proposed in the source literature, and a literature-derived behavioral effect size on a common Cohen's-d scale with a plausible range. Each participating dimension was coded on the 11 sensorimotor–affective dimensions described in the previous section; cosine distance between these profiles quantifies dimensional remoteness between paired modalities, and the landscape was visualized with UMAP (McInnes et al., 2018). Because both the dimension profiles and the effect sizes are coded from the literature rather than measured, this analysis is a quantitative synthesis of published claims, not new measurement.

Results. This analysis contextualizes the preceding ones. Sensorimotor distance correlates negatively with behavioral effect size across the 34 pairings ($r = -.379$, $p = .027$): the further apart two modalities are in sensorimotor space, the weaker the correspondence between them. Pitch and hue have one of the largest pairwise sensorimotor distances in the auditory–visual CMC literature, and pitch–hue occupies an isolated peripheral position in the UMAP embedding. Its weakness relative to pitch–elevation or loudness–size therefore carries theoretical information rather than being an accident: the modalities share no direct prothetic dimension, so the correspondence must be brokered by higher-order structure.

4 Discussion

4.1 Is There Structure in Three Centuries of Disagreement?

Yes, but the structure is family-specific and affectively mediated rather than universal or physically determined. Global agreement is low (mean 0.22), confirming that no pitch–hue mapping commands cross-cultural authority. At the same time, hue assignments are non-uniformly distributed (Rayleigh $R = 0.66$, $p = .003$), the Pedagogical and Synesthetic families sit above the flat baseline while the Cosmological family sits below it, and this ordering-though not individually significant at current family sizes-is coherent across every way we sliced the corpus.

The central methodological contribution of this paper is establishing that the reasoning family, not the individual system, is the correct unit of analysis for the historical corpus. All prior studies computing "agreement" have pooled heterogeneous traditions and observed global disagreement, thereby missing the structure hiding within each tradition. The eight-family classification dissolves the apparent contradiction between "correspondences exist" (within families, they do) and "no consensus exists" (across families, there isn't one). This is meta-analytic heterogeneity in its classic form: pooling across subgroups with different underlying mappings masks the agreement present within each subgroup. The Cosmological family sharpens the point from the other direction-its within-tradition agreement falls *below* the pooled baseline, so grouping by tradition does not merely recover hidden structure but also exposes traditions that carry none. One honesty constraint applies throughout: with 3–5 systems in most families, neither the individual family deviations nor the global heterogeneity statistic survives permutation testing (Supplementary Table S2), so the family ordering is an effect-size-level pattern awaiting confirmation on an expanded corpus, not an established difference.

4.2 Which Account Best Explains the Structure?

Table 6 collects the verdicts.

Structural account: partial support. The semitone–hue distance correlation ($r = .393$) and the below-chance consensus error of IRP and FRI establish that interval structure leaves a detectable imprint on historical color assignments. But the flagship prediction fails: perfect fifths separate by 80.7°, not 180°, and the interval–distance function saturates well before reaching the complementary region. Structural regularities constrain the ordinal relationship between interval size and hue distance-a partial scaffold-but cannot fix specific pitch-to-hue assignments. Only the spectrally anchored isomorphism theories (IRP, FRI) match the historical consensus; an isomorphism anchored elsewhere (SSC) or routed through intermediate perceptual dimensions (MPA,

Table 6: Summary of the four theoretical accounts against the evidence.

Account	Key prediction	Key statistic	Verdict
Structural	Interval geometry maps to hue geometry	$r = .393$; IRP 46.2° vs. 90° null; fifths at 80.7°, not 180°	Partial: ordering, not assignment
Semantic	Shared language carries the mapping	Timbral $\rho = .368$, $p = .007$; solfège $\rho = .019$	Contributing, entangled with affect
Statistical	Co-occurrence constrains color spacing	$r = -.94$ (interval size vs. variance); Rayleigh $R = 0.66$	Supported for large-scale structure only
Hedonic	Shared affect aligns the modalities	Procrustes disparity 0.559, $p = .027$; Evaluation dominant	Best supported

PDM, HEM) does not. The structural signal in the data is an isomorphism signal: narrow, but genuine.

Semantic account: contributing but insufficient. Language models encode significant pitch–hue geometry under timbral descriptor conditions ($\rho = .368$, $p = .007$). The active channel is shared *experiential* vocabulary-"warm", "bright", "cutting"-rather than arbitrary name co-occurrence, since solfège labels produce nothing ($\rho = .019$). The effect is genuine but incomplete: timbral language carries heavy affective content, so the geometry signal under timbral conditions is partly hedonic. The semantic and hedonic accounts cannot be fully separated here, which is itself informative-language and affect are closely tied in the pitch–hue domain.

Statistical account: specific support for large-scale structure. The $r = -.94$ correlation between interval size and between-system variance is the most tightly predicted result in the analysis, albeit one computed over the six distinct circular interval distances. Environmental statistics constrain global ordering (distant pitches get distant colors) but not fine-grained assignment. The account's explanatory scope is thus specific: it tells you how far apart the colors should be, but not what they should be.

Hedonic account: the best-supported. The EPA Procrustes disparity (0.559, $p = .027$) is the strongest single-account result across the four tests. Evaluation is the primary aligning dimension

($r = .74$), so auditory and visual dimensions are connected most reliably through their shared hedonic value. The item-level embedding evidence points the same way: pitch–color term proximity tracks shared valence–arousal profiles ($\rho = .139$). The family data fit this reading: Pedagogical systems converge because educators independently settle on what *feels* right to learners, without sharing a physical theory, and Synesthetic correspondences appear to tap the same affective architecture as non-synesthetic CMCs-an amplification of a signal that is already present, rather than a separate system. Palmer and colleagues (Palmer et al., 2013) showed that music–color associations are emotion-mediated; our results locate that mediation on the Evaluation (good–bad) axis, with Potency and Activity contributing only weakly.

Pulling these together, pitch–hue correspondences appear to rest on a *shared affective geometry*, partly scaffolded by structural regularities and environmental statistics, faintly echoed in language under timbral conditions, and not recoverable from any single first-principles map. The hedonic account is necessary but not sufficient: without the structural scaffolding, the data's interval–distance relationship has no explanation.

4.3 Positioning Pitch–Hue Among Cross-Modal Correspondences

Pitch–hue is among the hardest auditory–visual CMCs to explain, and the sensorimotor landscape analysis suggests why. Pitch and hue are dimensionally remote: they share no prothetic (intensive) dimension of the kind that directly grounds pitch–size, pitch–brightness, and loudness–size correspondences (Marks, 1978; Walsh, 2003). The $r = -0.379$ correlation between sensorimotor distance and behavioral effect size ($p = .027$) captures this: the further apart two modalities are in sensorimotor space, the weaker their correspondence. Pitch–hue sits at the far end of this tradeoff, so any correspondence between them has to be brokered by higher-order structure-affective valence, cultural co-occurrence, symbolic convention-which is what the present results show. The peripheral UMAP position of pitch–hue in the auditory–visual cluster then follows from the sensorimotor distance account rather than being an anomaly.

4.4 Implications for Neural Congruency Research

The historical analysis exposes a critical unresolved problem in the prior literature: no principled, data-driven criterion for pitch–hue congruency exists that is independent of behavioral report. The field has been defining congruency by asking participants what feels congruent - and no neural correlate of pitch–hue congruency has been identified as a result. The present findings break this circularity directly. Because no single historical map commands ground-truth authority, congruency definitions for neuroimaging experiments must not be derived from specific historical systems such as Newton's or Scriabin's. Doing so imports an arbitrary cultural choice into the experimental design and biases RSA matrices against the actual structure in the data.

The right approach is experimentally defined congruency - for example, audiovisual Stroop paradigms in which participants hear a tone while viewing a written note name that either matches or conflicts with the sounded pitch. The EPA results support this choice precisely: the Stroop conflict manipulation creates a mismatch on the Evaluation dimension that this analysis identifies as the primary affective channel of the pitch–hue correspondence. Beyond the Stroop, the theoretical RDMs derived from IRP, FRI, and EPA alignment can be submitted directly to RSA against EEG or fMRI embeddings. For the first time, a neural investigation of pitch–hue processing has a quantitatively grounded, theory-derived model to test.

4.5 Limitations and Future Directions

Several limitations constrain the current conclusions, and we state them plainly because they bound what the analysis can and cannot support. First, and most important, the family-level results rest on small samples. Several analyzed families contain only a handful of systems; each family score carries a bootstrap interval and a label-permutation test (Supplementary Table S2), and none of the family deviations-nor the global heterogeneity statistic ($p = .81$)-reaches significance under permutation. The apparent ordering of families is therefore suggestive rather than established, and replication on an expanded catalogue is needed before the family ranking can be treated as settled. Second, the study tests a large battery of hypotheses - six interval hypotheses, six structural theories, ten corpus-level predictions, and multiple model–descriptor combinations - so individually significant results must be read in the context of this wide inferential surface. We designated four primary claims in advance of the confirmatory write-up (family heterogeneity, the EPA Procrustes alignment, the interval–variance correlation, and the timbral embedding alignment) and collect all primary tests with Benjamini–Hochberg FDR-corrected q-values in Supplementary Table S7. Of the four primary claims, three survive correction at $q < .05$ (the EPA alignment, $q = .045$; the interval–variance correlation, $q = .007$; the timbral embedding alignment, $q = .024$), as do the semitone–hue correlation, the Rayleigh non-uniformity, the agreement–consistency correlation, and the effect-size–distance relationship; the family-heterogeneity permutation tests do not ($q \geq .13$). A pre-registered confirmatory study would nonetheless put the surviving effects on firmer footing. Third, color standardization introduces measurement error that is difficult to quantify: resolving free-text and pigment-name specifications to CIELAB via crowd-sourced and reference databases is principled but noisy, and some of the low global agreement may reflect conversion noise rather than genuine disagreement. Fourth, the hedonic test operates on author-coded constructs at both ends: the 11-dimensional sensorimotor profiles of the cross-modal dimensions were coded by the authors on an inventory modeled on the Lancaster norms rather than taken from participant ratings, and the EPA coordinates of the ten aligned dimensions were assembled from heterogeneous published norms. The Procrustes result is significant under permutation, but both configurations are inferential constructs over a small term set ($n = 10$), and the strength of the result should be weighed against this construct uncertainty; replacing the coded profiles with participant-normed ratings is a direct next step. Fifth, the catalogue is not a random sample: systems that were never published, or specified colors in forms that could not be standardized, are excluded, and the retained systems skew toward formally specified, Western-centric proposals; the family agreement scores are therefore lower bounds on within-family coherence for well-documented traditions, not estimates of a universal population parameter. The EPA norms are likewise predominantly Western, so the Evaluation-first finding should be tested cross-culturally before being generalized. Finally, the sentence-transformer probing demonstrates geometry alignment rather than specific mapping prediction, and a positive timbral result does not rule out that the signal is driven by affective content in the descriptors rather than purely linguistic co-occurrence; the hedonic and semantic accounts overlap here, and disentangling them requires targeted descriptor construction.

These limitations also locate the contribution. This is a meta-analytic and methodological study - a curated catalogue, a reference-free agreement framework, and a direct comparison of accounts-rather than a behavioral experiment, and it should be read as one. Its purpose is to extract the structure latent in the historical record and to hand the field a quantitatively grounded, theory-derived target for the experiments that must follow. The three most important next steps are: (1) extend the catalogue through non-Western archival sources and non-published practitioner traditions, and report interval estimates on the expanded sample, to reduce both sampling bias and statistical fragility; (2) collect large-scale behavioral pitch–hue matching data across multiple cultural backgrounds to test whether the family-level historical structure predicts individual differences; and (3) deploy the IRP, FRI, and EPA-aligned theoretical RDMs as predictors in RSA of EEG and fMRI responses, testing directly whether neural geometry mirrors the structure identified here.

4.6 Conclusion

The historical record is a dataset encoding structure: pitch and hue are connected through a shared affective geometry, scaffolded by interval regularities and environmental statistics, with no single first-principles map capturing the full assignment. The congruency framework, family-agreement structure, and theoretical RDMs derived here give behavioral and neural research the first quantitatively grounded targets-IRP, FRI, and the EPA-aligned RDM-for the experiments that must follow.

Supplementary Material

The Supplementary Material (`supplementary_material.pdf`) comprises: **S1**, the full 70-system catalogue (system, year, reasoning family, retention status, exclusion reason); **S2**, family agreement bootstrap intervals, label-permutation p-values, the global heterogeneity test, and the simulated chance distribution of the agreement score; **S3**, comparison of the Pedagogical (F8) consensus mapping against canonical spectral order; **S4**, the full 3×6 sentence-transformer probing grid for both descriptor sets, with FDR-corrected q-values; **S5**, exact Procrustes permutation statistics and dimension-wise alignment; **S6**, the coded CMC database with effect sizes and key sources; **S7**, the consolidated multiplicity table of all primary tests; **S8**, mean hue distance by interval (Table); **S9**, full 10-hypothesis corpus-level battery (Table); **S10**, per-family harmonic structure (Figure); **S11**, pairwise theory alignment heatmap (Figure); **S12**, predicted hue wheels for the six structural theories overlaid on the consensus (Figure); **S13**, per-model predicted hue maps — MiniLM, MPNet, Nomic (Figure); **S14**, cross-linguistic naming and emotion-mediation control panels (Figure); **S15**, complete per-system mapping panel organised by reasoning family (Figure).

Open Practices Statement

The historical catalogue, CIELAB conversion pipeline, CIEDE2000 agreement code, distance matrices, and family classification codes are available at `https://github.com/alexTrod/pitch-hue-correspondences`. Full analysis scripts for the EPA Procrustes alignment, sentence-transformer probing, IRP/FRI evaluation, and FDR correction are available from the corresponding author upon reasonable request. None of the analyses were preregistered.

Conflict of Interest

The authors declare no conflicts of interest.

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Ethics Statement

This study involved no human participants, animal subjects, or personal data. No ethics approval was required.

Informed Consent

Not applicable.

Author Contributions

Alex Noah Feldman: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. Ercan Engin Kuruoglu: Conceptualization, Supervision, Writing – review & editing.

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